

SMITH ISLAND, WA, USA

WMO: 994180

Lat: 48.3206N

Lon: 122.8310W

Elev: 15

StdP: 101.14

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.1	0.7	N/A	N/A	N/A	N/A	N/A	N/A	21.7	7.6	19.9	7.6	6.3	20	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	18.8	N/A	17.4	N/A	16.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	16.4	14.3	DB	-2.5	23.1	2.7	1.6	-4.5	24.2	-6.1	25.2	-7.6	26.1	-9.6	27.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	6.3	6.3	7.5	9.1	10.8	12.4	13.6	13.9	13.1	10.6	7.9	6.2
	DBStd	3.39	2.46	2.23	1.92	1.53	1.39	1.41	1.51	1.28	1.28	1.50	2.23	2.44
	HDD10.0	538	116	104	79	35	7	0	0	0	0	11	67	119
	HDD18.3	3102	373	337	335	278	233	177	146	138	156	238	313	377
	CDD10.0	477	1	0	2	7	32	73	113	121	94	30	4	1
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	6.6	6.3	6.3	5.8	5.3	5.3	5.2	4.6	4.0	5.0	6.7	7.0
Precipitation	PrecAvg	626	77	51	58	48	46	35	20	23	29	60	97	78
	PrecMax	777	114	117	81	82	90	61	53	79	67	115	192	136
	PrecMin	474	28	8	25	8	11	6	3	3	3	19	29	34
	PrecStd	88	20	21	17	16	21	16	12	19	19	26	41	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	11.8	13.7	14.6	16.7	20.2	21.7	20.4	18.6	15.7	13.5	12.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	10.8	10.4	11.6	12.9	14.8	17.3	19.0	18.2	17.2	14.2	12.3	10.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	10.0	9.6	10.7	11.9	13.6	15.3	17.1	17.0	16.1	13.4	11.4	9.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	9.3	8.9	9.9	11.1	12.8	14.2	15.7	15.9	15.2	12.6	10.6	9.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.7	3.0	3.6	3.7	3.6	4.0	4.5	4.6	4.3	3.1	2.9	2.6
		MCDBR	3.4	3.4	4.4	4.8	4.9	5.7	6.7	6.2	5.6	3.9	3.4	3.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.318	0.316	0.328	0.350	0.374	0.355	0.346	0.351	0.339	0.329	0.323	0.314
	taud		2.531	2.554	2.536	2.467	2.415	2.488	2.544	2.530	2.561	2.575	2.526	2.497
	Ebn at Noon		762	854	896	902	889	906	910	891	872	825	746	713
	Edh at Noon		61	73	86	102	112	104	97	95	84	71	61	56
All-Sky Solar Radiation	RadAvg		0.94	1.82	2.83	4.26	5.44	5.95	6.52	5.51	3.94	2.17	1.12	0.76
	RadStd		0.13	0.24	0.33	0.40	0.43	0.57	0.46	0.36	0.33	0.24	0.17	0.10

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page